

# AUDITING GRADIENT RETENTION UNDER HIDDEN CONDITIONS: EXACT CANCELLATION, OBJECTIVE-DEPENDENT ALIGNMENT, AND THE LIMITS OF KAPPA

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## ABSTRACT

Learning systems often average gradients over hidden conditions: annotator groups with conflicting preferences, clients with different objectives, teammates with unobserved roles. It is unclear when such averaging merely aggregates useful signal and when it systematically suppresses it. We ask what survives the averaging, and how to measure it. We show that gradient cancellation is governed by more than the presence of hidden heterogeneity. In an analytically tractable mirror model, elementwise parameter-independent weight fields cancel exactly under a symmetric condition mixture, while a field with a differentiable baseline can retain a nonzero component away from uniform policies. For  $K$ -way mixtures, retention depends on the direction of the condition mixture in the probability simplex even at fixed distance from uniform, and objective-induced gradient channels can interfere, making retention non-monotone in the entropy weight. We separate these structural effects from finite-sample noise with a measurement protocol: deterministic rollouts can collapse retention into a weight ratio, and episode-noise readouts can understate structural retention by an order of magnitude. Under a single bounded definition, value fields align with hidden conditions in Overcooked and 510K while the plain policy gradient remains near-orthogonal (a  $1.9\times$  value-policy gap, not the  $30\text{--}65\times$  suggested by noisy readouts); advantage weighting with a differentiable baseline improves retention only modestly over the plain gradient. A controlled capacity experiment shows that severe cancellation can coexist with poor worst-condition performance; in contrast, on a real three-group preference dataset the condition-contrast energy is dominated by item-level noise and group-conditioned capacity yields no gain. In a switching task, retention need not predict return when the measured condition is not the variable the task relies on. Retention is a structural audit of the specified condition signal, not a general performance metric, and it indicates when condition-aware capacity is likely worth its cost.

## 1 INTRODUCTION

Supervised and reinforcement learning systems are routinely trained on data that mixes heterogeneous subpopulations. The subpopulation label is frequently missing: a preference dataset pools annotators with conflicting tastes, a federated client population encodes different task distributions, a multi-agent episode hides which role the partner plays. The model sees one input distribution, while the label or reward is generated by a condition that is not present in the input. Training then optimizes the mixture objective, and the update direction is the mixture of condition-specific gradients,

$$\bar{g} = \sum_i p_i g_i, \quad (1)$$

where  $g_i(\theta) = \nabla_{\theta} L_i(\theta)$  is the population gradient of the condition- $i$  objective under a fixed rollout distribution.

A mixture can erase signal. The extreme case is cancellation: if two conditions demand opposite behavior on the same input, their gradients can point in opposite directions and the average update is zero even though each condition carries a strong, learnable signal. Practitioners observe this as training that stalls without any visible loss anomaly. This paper asks how to audit that failure precisely: when cancellation is exact, when it is partial, how it depends on the objective, and what the resulting measurement does and does not tell us.

A companion phenomenon was reported previously: hidden relational variables can contract policy-gradient updates while value-based updates appear to survive. That report left the mechanism open. Our starting point is that the mechanism cannot be recovered from algorithm-family labels, because a gradient field is a property of the *objective evaluated on a fixed policy and data*, not of the optimizer. We therefore fix the measurement basis: same network, same rollout distribution, same condition definition, and change only the scalar objective whose gradient defines the field.

**Thesis.** When hidden conditions are averaged, what survives is determined not only by symmetry between conditions, but by the geometry of the condition mixture and by the objective-induced structure of the gradient field. The results below are chapters of that one question.

## Contributions.

1. **Structure: which fields survive, and why.** (a) For the mirror bandit, any elementwise, parameter-independent weight field satisfies  $\mathbb{E}[g_B] = -\mathbb{E}[g_A]$ , hence zero shared gradient under the symmetric mixture; the same normalization cancels  $K$ -way matching mixtures at every  $K$ . (b) The soft-Q field has an exact closed form,  $\kappa_{\text{mix}} = \Delta^2/(\Delta^2 + 4r^2)$  with  $\Delta = \alpha \log(\rho/(1-\rho))$ , verified against autograd to machine precision. (c) The advantage-weighted field with a differentiable baseline is the only policy-gradient-style field that survives the symmetric mixture, through an implicit even coupling. (d) Direction dependence appears at  $K \geq 3$ , and entropy and mixture channels interfere with a computable minimum. These are properties of the condition simplex, not of a non-abelian group.
2. **Measurement: what a retention score can and cannot be.** The bounded mixture-reference retention separates structural retention from episode noise. Deterministic rollouts reduce the score to a single-action weight ratio on the mirror model (0.633 where the correct protocol gives 0.001), and episode-noise readouts can sit an order of magnitude below structural scores. Two fields must be compared on the same episode batch, and condition-contrast energy must be read against within-condition noise before a conclusion is drawn.
3. **Boundaries: when the audit recommends capacity.** Under one bounded definition, value fields align with hidden conditions (Overcooked dynamic 0.998, 510K 0.95–0.98) while the plain policy gradient remains near-orthogonal (0.529); advantage weighting with a differentiable baseline improves retention only modestly (0.590), and the value–policy gap is about  $1.9\times$ , not the  $30\text{--}65\times$  implied by noisy readouts. A controlled capacity experiment shows that severe cancellation with contrast energy dominating noise is fixed by condition-aware capacity (worst condition 0.003 to 1.97), whereas a real three-group preference dataset with contrast dominated by item-level noise gains nothing from conditional capacity. A switching task shows retention need not predict return when the measured condition differs from the variable the task relies on. The audit recommends intervention only when low retention coincides with contrast energy above noise and a task-relevant condition.

## 2 RELATED WORK

**Gradient conflict and multi-task optimization.** PCGrad (Yu et al., 2020), CAGrad (Liu et al., 2021), MGDA (Sener & Koltun, 2018), GradNorm (Chen et al., 2018), GradDrop (Chen et al., 2020), and later gradient-surgery methods treat conflict as an observed quantity to be projected, reweighted, or scheduled away. We study the complementary question: when conflict is *structural*, i.e. forced by hidden-condition symmetry before any optimizer intervenes. Our audit is a diagnostic, not another surgery rule, and we do not claim to replace those methods. The retention score is not a cosine similarity: for two equally weighted conditions it reduces to a monotone function of

Table 1: Measured fields. Every entry is a gradient with respect to the same policy parameters on the same rollouts; only the scalar objective changes.

| Field     | Objective or weight                                             | Note                  |
|-----------|-----------------------------------------------------------------|-----------------------|
| reinforce | $\mathbb{E}_a[\nabla(-\log \pi(a) Q_r(a))]$                     | parameter-free weight |
| expq      | $\nabla(-\sum_a \pi(a) Q_r(a))$                                 | full objective        |
| softq     | $\nabla \sum_a \pi(a)(\alpha \log \pi(a) - Q_r(a))$             | detached $Q$          |
| awr       | $\mathbb{E}_a[\nabla(-\log \pi(a) e^{(Q_r(a)-V)/\tau})]$        | differentiable $V$    |
| softmaxq  | $\mathbb{E}_a[\nabla(-\log \pi(a) \text{softmax}(Q_r/\tau)_a)]$ | $Q$ -dependent weight |

the gradient cosine only when the two gradient norms match, and the mixture-reference ratio stays defined when the average cancels and extends to  $K$  conditions with general weights.

**Hidden context and preference aggregation.** Distributional preference learning (Siththaranjan et al., 2024) shows that averaging unobserved annotator groups distorts a reward model; heterogeneous-preference alignment work studies minority under-representation and equitable aggregation (Chakraborty et al., 2024). We work in gradient space rather than risk or reward space and provide exact cancellation conditions plus objective-specific survival rates; we do not propose a new alignment algorithm.

**Symmetry, averaging, and invariance.** Averaging over a group acts as a projection onto invariant components (a Reynolds operator), and the resulting decomposition is classical. Our use of it is instrumental: we apply it to condition-conditioned gradient fields and derive measurement consequences. We explicitly do not claim group averaging as a contribution.

**The companion report.** A prior study (citation withheld for double-blind review) introduced the retention ratio and reported estimator dependence and an apparent policy-gradient/value reversal. That report is a phenomenon study. The present work replaces the algorithm comparison with a common-basis field measurement, derives the closed forms, and corrects two measurement practices on which the earlier scores depended.

### 3 PROBLEM SETUP AND MEASUREMENT

#### 3.1 HIDDEN-CONDITION GRADIENT FIELDS

Let  $i$  index a hidden condition with mixture weights  $p_i$ , and let  $g_i = \nabla_{\theta} L_i$  denote the population condition gradient of a scalar objective  $L_i$  under a fixed rollout distribution. We report the bounded structural retention

$$\kappa_{\text{mix}} = \frac{\|\bar{g}\|^2}{\sum_i p_i \|g_i\|^2}, \quad 0 \leq \kappa_{\text{mix}} \leq 1, \quad (2)$$

where the upper bound follows from convexity. We also report the energy decomposition

$$E_{\text{shared}} = \|\bar{g}\|^2, \quad E_{\text{contrast}} = \sum_i p_i \|g_i - \bar{g}\|^2, \quad E_{\text{mix}} = \sum_i p_i \|g_i\|^2. \quad (3)$$

The historical uniform-reference normalization  $\|\bar{g}\|^2 / \text{mean}_i \|g_i\|^2$  is not bounded for non-uniform  $p$  and is used only to read legacy result files.

#### 3.2 FIELDS UNDER STUDY

All fields are gradients with respect to the same policy parameters on the same data (Table 1).

The distinction between a differentiable and a detached baseline matters: only the former changes the parity structure of the field, so we refer to our awr as an advantage-weighted field with a differentiable baseline rather than as standard AWR.

### 3.3 MEASUREMENT PROTOCOL

Three rules follow from the audit in Section 5.2 and are used throughout. (i) Rollouts are stochastic and policy-weighted; deterministic (argmax) rollouts collapse the score into a weight ratio on the mirror model. (ii) Conditions are forced per rollout when the environment allows it, so the measured contrast is between clean condition assignments rather than between random mixes. (iii) When two fields are compared, both are computed from the *same* episode batch. We report structural scores from condition means and, separately, the one-episode noise diagnostic

$$\kappa_{\text{ep}} = \frac{E_{\text{shared}}}{E_{\text{mix}} + \sum_i p_i \sigma_i^2}, \quad (4)$$

where  $\sigma_i^2$  is the within-condition episode-gradient variance. A finite-sample mean calibration,  $E_{\text{shared}}/(E_{\text{mix}} + \sum_i p_i \sigma_i^2/n_i)$ , is reported as a surrogate and is not a high-probability guarantee.

## 4 THEORY

### 4.1 EXACT CANCELLATION

**Proposition 1** (Mirror cancellation). *In the two-action mirror bandit with  $Q_A = (r, -r)$  and  $Q_B = (-r, r)$ , any field whose weight is elementwise in  $Q_r$  and independent of the policy parameters,  $w_r(a) = f(Q_r(a))$ , satisfies  $\mathbb{E}[g_B] = -\mathbb{E}[g_A]$ . Under the symmetric mixture  $p = (1/2, 1/2)$ ,  $\mathbb{E}[\bar{g}] = 0$  and  $\kappa_{\text{mix}} = 0$  exactly.*

*Verification.* Closed form and autograd agree at the  $10^{-32}$  level for the stored configurations, covering reinforce, expq, awr without baseline, and softmaxq (data/kappa/toy\_fields/ol\_closed\_forms.json).

**Proposition 2** ( $K$ -way normalization). *In the  $K$ -way matching bandit with  $Q_i(b) = 2R \delta(i, b)$  and a uniform condition mixture,  $\sum_i g_i = 0$  for every  $K$ ; the cancellation is a normalization identity, not a group-theoretic accident. Verified for  $K = 2, 3, 4$  with  $\kappa_{\text{mix}} < 1.8 \times 10^{-32}$ .*

The boundary is explicit: the result concerns expected gradients under the matching construction. Real environments are not exact mirrors, and the measured structural scores there are around 0.5, i.e. near-orthogonal rather than cancelling.

### 4.2 CLOSED FORM FOR THE SOFT-Q FIELD

**Proposition 3** (Soft-Q retention). *In the mirror bandit, let  $\rho = \pi(\text{action } 0)$ ,  $q = 1 - \rho$ , and  $\Delta = \alpha \log(\rho/q)$ . The soft-Q field satisfies*

$$c_A = \rho q(\Delta - 2r), \quad c_B = \rho q(\Delta + 2r), \quad g_i = c_i u, \quad \kappa_{\text{mix}} = \frac{\Delta^2}{\Delta^2 + 4r^2}. \quad (5)$$

*Verification.* Closed form versus autograd of the explicit expected loss agrees to  $5.0 \times 10^{-16}$ ; Monte Carlo anchors agree within sampling error; the stored  $\alpha$ -grid values match to the archived precision (one archived point differs by 0.001 under a documented old rounding convention; verify\_closed\_forms\_ol.py). Limiting cases:  $\alpha \rightarrow 0$  degenerates to the cancelling expected- $Q$  field, while a non-uniform policy or a larger entropy weight increases retention. This is a fixed-policy identity; it does not describe a training trajectory.

The advantage-weighted field with a differentiable baseline is the only policy-gradient-style field that survives the symmetric mixture: the baseline enters the weight through  $V = \sum_a \pi(a)Q(a)$  and creates an implicit even coupling. Its closed form appears in Appendix A; at uniform policies it also cancels for every temperature, and for non-uniform policies retention is non-monotone in  $\tau$ .

### 4.3 $K$ -WAY GEOMETRY AND CHANNEL INTERFERENCE

**Proposition 4** (Direction dependence). *For the  $K$ -way matching field,  $g_i[b] = -2R \pi_b(\delta(i, b) - \pi_i)$  and  $\bar{g}[b] = -2R \pi_b(p_b - \langle p, \pi \rangle)$ , where the second term is the projection of the mixture onto the policy-orthogonal complement. When the condition simplex has dimension at least two ( $K \geq 3$ ), retention at fixed  $L_1$  distance from the uniform mixture varies with direction.*

Measured spread at  $L_1$  distance 0.45 (200 directions):  $K = 3$ :  $\kappa_{\text{mix}} \in [0.071, 0.219]$  ( $3.10\times$ );  $K = 4$ :  $[0.051, 0.210]$  ( $4.08\times$ ). This is simplex geometry; any claim that the effect is specific to  $S_3$  or to non-abelian symmetry is withdrawn. Direction dependence is absent for  $K = 2$  because the mixture simplex is one-dimensional.

**Proposition 5** (Channel interference). *The soft-Q field is the sum of an entropy channel and a condition-mixture channel in the same weighted tangent space. When these two directions have positive weighted inner product, retention is non-monotone in  $\alpha$ : the numerator cancels before the ratio recovers.*

Measured minima:  $\kappa_{\text{mix}} = 7.8 \times 10^{-4}$  at  $\alpha = 1.78$  for  $K = 2$  with  $p = (0.8, 0.2)$ , and  $1.1 \times 10^{-3}$  at  $\alpha = 1.41$  for  $K = 3$  with  $p = (0.7, 0.2, 0.1)$ . The numerator cancellation point and the ratio minimum are distinct and must be reported separately.

A dynamical reading of this interference was proposed previously and is withdrawn here. Under the implemented objective  $L = \sum_a \pi(a)(\alpha \log \pi(a) - Q(a))$ , gradient descent on the symmetric mixture minimizes  $-\alpha H(\pi)$  and therefore moves toward higher entropy, not toward a peaked policy. Static channel interference survives as a statement about the field, not about an attractor.

## 5 EXPERIMENTS

All numbers are produced by the canonical implementation and collected in `notes/canonical_metric_results.md`; raw files are listed in Appendix B.

### 5.1 EXACT BANDIT: CANCELLATION AND CLOSED FORMS

On the mirror bandit we evaluate one policy with identical parameters and an identical evaluation seed schedule under two observation protocols: revealed (the partner one-hot is part of the observation) and masked (the partner slot is zeroed, so the relation is hidden while the reward and the analytic  $Q$  still depend on it). Under masking, `expq` cancels exactly ( $\kappa_{\text{mix}} = 0$ ,  $E_{\text{shared}} = 0$ ), `reinforce` is indistinguishable from zero (0.0025), `softmaxq` and `softq` retain 0.0005 and 0.0045, and the differentiable-baseline `awr` field retains the largest small component (0.0091); the soft-Q component follows the closed form and grows with  $\alpha$  (Proposition 3). Under the revealed protocol all five fields collapse to the same value (0.4397–0.4402), the expected degeneracy of a policy that already solves the matching task. Because parameters, rollouts, and seeds are fixed and only the observation mask changes, the contrast isolates condition visibility (`data/kappa/toy-fields/visibility_control.json`); five random-init policies of the same architecture show the same pattern.

### 5.2 PROTOCOL AUDIT: DETERMINISTIC ROLLOUTS INVENT A FIELD AXIS

On the hidden mirror bandit, deterministic (`argmax`) rollouts visit a single action per condition, and every field reduces to the ratio of weights at that action. Measured artifacts: `softmaxq` reports 0.633 under deterministic rollouts and 0.0011 under the correct stochastic protocol; `awr` reports 0.551 versus 0.028. `expq` remains exactly zero under both protocols, and the revealed-policy values are protocol-robust ( $\sim 0.396$  both ways). The audit rule is direct: an apparent field axis measured with `argmax` rollouts on a degenerate model should be treated as a weight ratio until recomputed with policy-weighted sampling.

### 5.3 REAL ENVIRONMENTS UNDER THE CANONICAL METRIC

**Overcooked (switching-preserving protocol).** One episode batch per checkpoint is shared by all fields, and only the measured objective changes, so the comparison is paired by construction (Table 2). The value field’s condition-mean gradients nearly coincide on the dynamic model (0.998). Among policy-gradient-style fields the differentiable-baseline `awr` field has the highest retention (0.590), but it stays far below the value field and only modestly above the plain policy gradient (0.529): the large policy-gradient/value separation of the mirror model does not transfer here as a corresponding advantage-field separation. Here `awr` uses the observed return-to-go as the advantage sample and the value head as the differentiable baseline. Under the start-partner protocol the same

Table 2: Overcooked structural retention (three seeds). All fields share one episode batch per checkpoint, and the `awr` entry keeps the value baseline in the autograd graph. The value entry is the diagnostic field  $-\sum_t V(s_t)$ ; it is not a TD residual.

| Field                            | static            | dynamic                             |
|----------------------------------|-------------------|-------------------------------------|
| <code>reinforce</code>           | $0.515 \pm 0.021$ | $0.529 \pm 0.034$                   |
| <code>awr</code>                 | $0.500 \pm 0.000$ | $0.590 \pm 0.021$                   |
| <code>value</code> ( $-\sum V$ ) | $0.568 \pm 0.027$ | <b><math>0.998 \pm 0.000</math></b> |

Table 3: 510K structural retention, paired protocol (six seeds per level). Both fields are computed from the same episode batch.

| Visibility $p$ | <code>reinforce</code> | <code>value</code>                  |
|----------------|------------------------|-------------------------------------|
| 0.0 (hidden)   | $0.514 \pm 0.099$      | <b><math>0.980 \pm 0.011</math></b> |
| 0.5            | $0.514 \pm 0.084$      | <b><math>0.974 \pm 0.018</math></b> |
| 1.0 (visible)  | $0.513 \pm 0.092$      | <b><math>0.948 \pm 0.017</math></b> |

conclusion appears at the level of condition means (static 0.533, dynamic 0.502), and a memory intervention does not change it (m4 0.498, m8 0.505, m16 0.500): supplying memory does not restore alignment, a negative intervention result.

**510K (paired protocol).** Using checkpoints trained with hidden teammate visibility  $p \in \{0, 0.5, 1\}$  and computing both fields from the same episode batch, per-step reward-weighted policy gradients retain 0.514/0.514/0.513 while value fields retain 0.980/0.974/0.948 (Table 3). The ordering is stable at every seed; the visible/hidden difference is small and we do not interpret it as a trend.

**Reading.** Across the two environments the value fields align with hidden conditions, while the policy-gradient-style fields sit near 0.5 (orthogonal; in Overcooked the differentiable-baseline advantage field is the least cancelled, 0.590). Expressed as a value-policy ratio this is about  $1.9\times$ . The previously reported  $30\text{--}65\times$  separations came from comparing one-episode noise scores ( $\kappa_{ep}$ , where the value field’s within-condition variance is small and the policy-gradient field’s variance is large) rather than structural scores. We report both quantities, but only the structural one supports field-level statements.

#### 5.4 CAPACITY, NOT OPTIMIZER

On an exact  $K$ -way training comparison with a shared policy versus a condition-aware policy (full-batch, deterministic, same initial logits), the shared model’s limitation is a mixture compromise (Table 4). With a uniform mixture and the linear objective, the shared model stays at average reward 0.667 while the conditional model reaches 1.99 ( $\kappa_{mix} = 0$  at initialization, by cancellation). Under mild and strong asymmetric mixtures the shared model improves the average (1.20 and 1.60) but leaves the worst condition at 0.003 and 0.002, while the conditional model keeps the worst condition at 1.97 and 1.93. Turning on the entropy channel ( $\alpha = 2$ ) trades average for worst-condition performance in the shared model (0.752 average, 0.517 worst), a Pareto movement rather than a fix. The intervention suggested by the audit is conditional information or capacity, not an arbitrary gradient rule.

#### 5.5 REAL THREE-GROUP PREFERENCE DATA

To test whether the audit extends beyond RL bandits, we use DICES-350 (Aroyo et al., 2023): 350 adversarial conversations, each rated by all 123 raters, with rater demographics; the dataset is released under CC BY 4.0. We treat a demographic attribute as the hidden condition, use the binary perceived-harm judgment (Yes vs No, Unsure dropped), split by `item_id` (245 train / 105 test

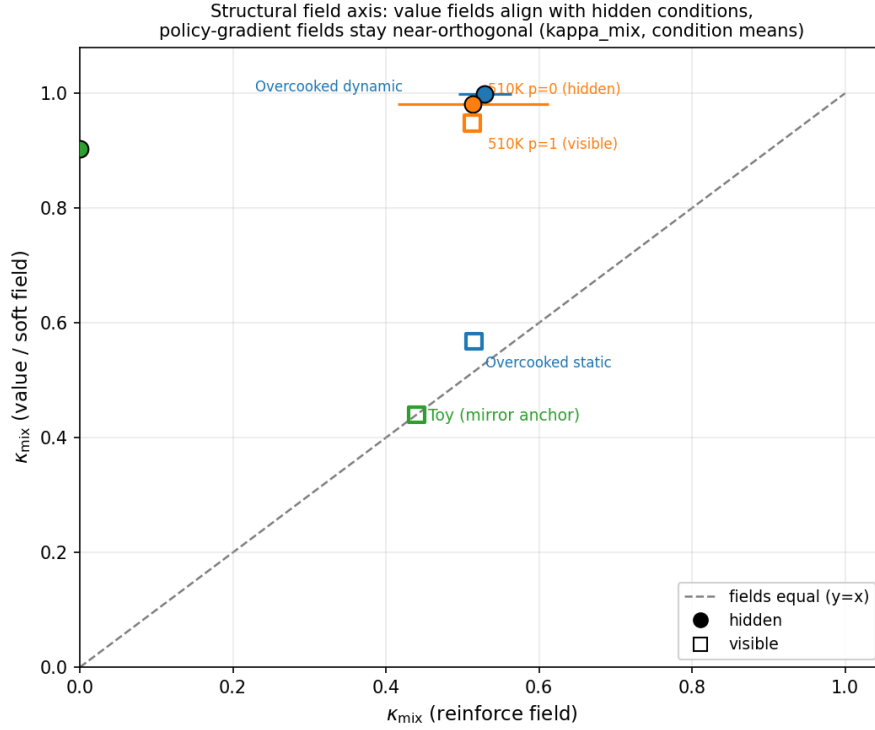


Figure 1: Structural field axis under  $\kappa_{\text{mix}}$  (condition means). Value fields sit above the diagonal; policy-gradient fields stay near 0.5 in the real environments. The mirror-bandit point is an exact anchor, not a real environment.

Table 4: Shared versus condition-aware capacity (exact full-batch training). Rewards are reported as average / worst condition.

| Mixture | Model                   | avg. reward | worst reward | initial $\kappa_{\text{mix}}$ |
|---------|-------------------------|-------------|--------------|-------------------------------|
| uniform | shared, $\alpha=0$      | 0.667       | 0.466        | 0.000                         |
| uniform | conditional, $\alpha=0$ | 1.992       | 1.992        | —                             |
| mild    | shared, $\alpha=0$      | 1.197       | 0.003        | 0.208                         |
| mild    | conditional, $\alpha=0$ | 1.991       | 1.970        | —                             |
| mild    | shared, $\alpha=2$      | 0.752       | 0.517        | 0.445                         |
| strong  | shared, $\alpha=0$      | 1.597       | 0.002        | 0.501                         |
| strong  | conditional, $\alpha=0$ | 1.991       | 1.933        | —                             |

items), and fit a TF-IDF plus logistic model shared across raters. Three seeds; the shared model reaches 0.73 train accuracy. Table 5 reports the audit and the capacity comparison.

On the race axis, the contrast energy is about six times smaller than the within-group item noise ( $E_{\text{contrast}}/\sigma^2 \approx 0.17$ ), and group-conditioned models do not improve average or worst-group accuracy. The audit therefore predicts that this is not a capacity problem, and the held-out comparison confirms it. This complements the bandit capacity experiment: there, contrast dominates noise and condition-aware capacity raises the worst condition from 0.003 to 1.97; here, contrast is noise-dominated and conditional capacity buys nothing. Caveats:  $Q_{\text{overall}}$  is perceived harm, not objective harm; `Unsure` labels are dropped; the model is a linear bag-of-ngrams classifier, and no group-DRO or reweighting baseline is included.

Table 5: DICES-350 supervised check (three seeds). The race axis carries the strongest condition signal, but its contrast energy is dominated by within-group item noise, and conditional capacity yields no held-out gain.

| Axis (groups) | $\kappa_{\text{mix}}$ | $E_{\text{contrast}}$ | $\sigma^2$ | shared / cond. avg-worst  |
|---------------|-----------------------|-----------------------|------------|---------------------------|
| race (3)      | $0.116 \pm 0.025$     | 0.0020                | 0.0113     | 0.656/0.609 / 0.654/0.607 |
| age (3)       | $0.033 \pm 0.005$     | 0.0003                | 0.0064     | 0.640/0.623 / 0.638/0.624 |
| gender (2)    | $0.001 \pm 0.000$     | 0.0005                | 0.0035     | 0.641/0.623 / 0.640/0.625 |

Table 6: Switching task: objectives trained from scratch under an identical budget. Post-switch reward is measured on steps 20–28 after the flip.

| Arm       | return                             | post-switch reward | final $\kappa_{\text{mix}}$ |
|-----------|------------------------------------|--------------------|-----------------------------|
| td        | <b><math>36.97 \pm 0.06</math></b> | 0.75               | $0.06 \pm 0.04$             |
| softq     | $28.62 \pm 11.81$                  | 0.58               | $0.94 \pm 0.09$             |
| reinforce | $9.48 \pm 5.13$                    | 0.18               | $0.43 \pm 0.18$             |
| awr       | $5.03 \pm 2.02$                    | 0.04               | $0.57 \pm 0.08$             |

## 5.6 RETENTION IS NOT A PERFORMANCE SCORE

In a switching hidden-matching task with four objectives trained from scratch under an identical budget (eight seeds), final returns and final structural retention are shown in Table 6. The arm with the lowest measured retention is the best performer. The explanation is variable mismatch: the task requires tracking a phase from reward history, which is orthogonal to the initial partner label that retention conditions on. The diagnostic answers “how much of the specified condition signal survives averaging,” not “will training succeed.” A low retention score must therefore be paired with a check that the condition is the variable the task actually relies on.

## 6 WHAT THE AUDIT DOES AND DOES NOT CLAIM

The retained statements are: (i) exact cancellation conditions on the mirror and matching models; (ii) an exact closed form for the soft-Q field and the differentiable-baseline exception; (iii)  $K$ -way direction dependence and channel interference as geometry; (iv) protocol and noise separation as methodology; (v) structural alignment of value fields versus near-orthogonality of the plain policy gradient in two environments, with the differentiable-baseline advantage field the least cancelled among the policy-gradient fields in Overcooked; and (vi) decoupling from performance. The operational reading is a conditional rule: when retention is low, contrast energy exceeds within-condition noise, and the measured condition is the variable the task relies on, condition-aware capacity is the intervention; when any of the three fails, the audit recommends no intervention. We explicitly do not claim that value methods are universally better, that policy-gradient fields always vanish, that direction dependence is group-specific, that retention predicts return, or that memory or entropy interventions fix cancellation.

## 7 LIMITATIONS

The exact results are bandit-level. Real-environment claims rest on condition-mean gradients under one measurement protocol and on small seed counts (three seeds for the Overcooked field axis). The value field is the diagnostic  $-\sum V$ , not a TD residual; its alignment is a property of the fitted value function on the measured distribution. The Overcooked awr entry uses the observed return-to-go as the advantage sample and the learned value head as the differentiable baseline, so it is the practical analogue of the analytic field; a dataset-level stop-gradient shift keeps the float32 exponential finite, which multiplies the field by a positive scalar and leaves  $\kappa_{\text{mix}}$  unchanged. The 510K environment has weak relational drive and is used as a supporting measurement, not as the primary demonstra-

tion. The real-data check uses a single dataset (DICES-350), a linear TF-IDF model, and binary perceived-harm labels; it includes no conditional-routing, group-DRO, or reweighting baseline, and replication on a second dataset is left to future work. The finite-sample analysis reports a deterministic surrogate; a high-probability guarantee for the ratio estimator would require an explicit distributional assumption.

#### REPRODUCIBILITY STATEMENT

The metric implementation is `src/iigc/metrics/kappa.py`; the closed forms, protocol and visibility controls,  $K$ -way controls, Overcooked and 510K measurements, and the capacity comparison are scripted under `experiments/common_basis/`. Result files carry a `measurement_metadata` block recording the gradient definition, condition weights, rollout protocol, parameter space, and noise treatment. Unit tests cover the closed forms, cancellation, boundedness, and the metadata contract. No human subjects or private data are involved. Full provenance is listed in Appendix B.

#### AI USE STATEMENT

In this work, we used generative AI tools for drafting assistance and code editing suggestions. We have reviewed and verified all AI-assisted work: all closed-form results were checked against independent autograd and Monte Carlo computations, and all reported numbers are regenerated by the scripts listed in Appendix B. We take responsibility for the final content of this work.

#### ETHICS STATEMENT

This work studies optimization diagnostics for models trained on mixtures of hidden subpopulations. It does not involve human subjects or private data. The discussed failure mode (minority conditions being under-represented by averaged objectives) has fairness implications; we present the audit as a diagnostic and make no deployment recommendation.

#### REFERENCES

- Lora Aroyo, Alex Taylor, Mark Diaz, Christopher Homan, Alicia Parr, Greg Serapio-Garcia, Ding Wang, and Gaurang Ghosh. Dices dataset: Diversity in conversational ai evaluation for safety. In *Advances in Neural Information Processing Systems: Datasets and Benchmarks Track*, 2023.
- Souradip Chakraborty, Jiahao Qiu, Hui Yuan, Alec Koppel, Furong Huang, Dinesh Manocha, Amrit Singh Bedi, and Mengdi Wang. Maxmin-rlhf: Alignment with diverse human preferences. In *International Conference on Machine Learning*, 2024.
- Zhao Chen, Vijay Badrinarayanan, Chen-Yu Lee, and Andrew Rabinovich. Gradnorm: Gradient normalization for adaptive loss balancing in deep multitask networks. In *International Conference on Machine Learning*, 2018.
- Zhao Chen, Jiquan Ngiam, Piyush Sapra, Qizhen Le, et al. Just pick a sign: Optimizing deep multitask models with gradient sign dropout. In *Advances in Neural Information Processing Systems*, 2020.
- Bo Liu, Xingchao Liu, Xiaojie Jin, Peter Stone, and Qiang Liu. Conflict-averse gradient descent for multi-task learning. In *Advances in Neural Information Processing Systems*, 2021.
- Ozan Sener and Vladlen Koltun. Multi-task learning as multi-objective optimization. In *Advances in Neural Information Processing Systems*, 2018.
- Anand Siththaranjan, Cassidy Laidlaw, and Dylan Hadfield-Menell. Distributional preference learning: Understanding and accounting for hidden context in rlhf. In *International Conference on Learning Representations*, 2024.
- Tianhe Yu, Saurabh Kumar, Abhishek Gupta, Sergey Levine, Karol Hausman, and Chelsea Finn. Gradient surgery for multi-task learning. In *Advances in Neural Information Processing Systems*, 2020.

## A DIFFERENTIABLE-BASELINE CLOSED FORM

For the advantage-weighted field with  $w = \exp((Q_r - V)/\tau)$  and a differentiable baseline  $V = \sum_a \pi(a)Q_r(a)$ , write  $x = 2r/\tau$ ,

$$A_0 = e^{xq}, \quad A_1 = e^{-xp}, \quad M_A = p \log p A_0 + q \log q A_1, \quad (6)$$

$$B_0 = e^{-xq}, \quad B_1 = e^{xp}, \quad M_B = p \log p B_0 + q \log q B_1, \quad (7)$$

with  $p = \pi(\text{action } 0)$ ,  $q = 1 - p$ . Then

$$c_A = pq(A_1 - A_0 + xM_A), \quad c_B = pq(B_1 - B_0 - xM_B). \quad (8)$$

At  $p = 1/2$  the field cancels exactly for every  $\tau$ ; otherwise retention has an interior maximum in  $\tau$ . Closed form and autograd agree to  $9.2 \times 10^{-16}$  and Monte Carlo anchors agree within three standard errors (`verify_closed_forms_ol.py`).

## B NUMBER PROVENANCE

Table 7: Provenance of the main-text numbers.

| Result                                         | File or script                                                    |
|------------------------------------------------|-------------------------------------------------------------------|
| Cancellation, soft-Q and AWR closed forms      | <code>data/kappa/toy_fields/ol_closed_forms.json</code>           |
| $K$ -way cancellation, direction, interference | <code>data/kappa/toy_fields/kway_geometry.json</code>             |
| Protocol audit (deterministic vs stochastic)   | <code>data/kappa/toy_fields/det_vs_stoch.json</code>              |
| Controlled visibility contrast (mirror bandit) | <code>data/kappa/toy_fields/visibility_control.json</code>        |
| Overcooked field axis (shared episode batch)   | <code>data/kappa/server_tasks/results/oc_field_axis.json</code>   |
| Overcooked switch and memory protocol          | <code>data/kappa/server_tasks/results/oc_switch_kappa.json</code> |
| 510K paired protocol                           | <code>data/kappa/server_tasks/results/510k_field_axis.json</code> |
| Shared vs conditional capacity                 | <code>data/kappa/compromise_training/results.json</code>          |
| Switching-task decoupling                      | <code>data/kappa/o4_adaptive/aggregate.json</code>                |
| DICES-350 supervised check                     | <code>data/kappa/dices350/audit.json</code>                       |
| Canonical metric implementation                | <code>src/iigc/metrics/kappa.py</code>                            |

## C CORRECTION LOG

Superseded claims and their replacements are recorded in `notes/paper1_propositions_corrections.md`, including the withdrawn soft-Q dynamics claim, the parameter-space scope of the Fisher/KL variant, and the replacement of  $S_3$ -specific language with  $K$ -way simplex geometry.